Faculty of Engineering & Technology
Computer Science Department

Database Systems
COMP333

Course Project

Phase 4

Prepared by:
Mohammad Abu Hijleh **ID: 1221350**
Laith Ghoneim **ID: 1221123**

Instructor: **Bassem Sayrafi**

Section: **2**
Date: **10/6/2025**

Table of Contents

Project Scope & Description:

The Cobra Shop Management System is a database-driven application designed to streamline the operations of Cobra Shop a computer and accessories store located in Ramallah. The system aims to enhance efficiency in multiple aspects of the business, including:

- Sales Transactions: The system will manage point-of-sale transactions, invoice generation, and tracking of customer purchases.
- Inventory Tracking: Real-time stock updates, low-stock alerts, and warehouse management to ensure proper stock levels.
- Supplier Management: Keeping records of suppliers, their supplied products, and restocking schedules.
- Customer Management: Storing customer details, purchase history, and enabling loyalty programs or discounts.
- Reporting & Insights: Generating detailed reports on sales trends, best-selling products, and revenue analysis to support business decision-making.

By implementing this system, Cobra Shop will improve its operational efficiency, minimize manual errors, and enhance customer satisfaction through seamless service.

Entities and Relationships:

The database system will consist of many entities with relationships between them:

Entities:

1. **Staff** *(Staff_ID, Manager_ID, Name, Position, Section_ID, Salary, Hire_Date, Gender, Age, Assignment_count)* - Employees working at the shop.
2. **Customers** *(Customer_ID, Name, Email, Phone, Address, Birth_Date)* - Information on customers who make purchases.
3. **Products** *(Product_ID, Category_ID, Title, Price, Stock, Description, image_path)* - Items available in the shop.
4. **Suppliers** *(Supplier_ID, manager_id, Name, Contact, Email, Address)* - Details of suppliers who provide products.
5. **Warehouse** *(Warehouse_ID, manager_id, Product_ID, Location_Name, Stock_Quantity, Last_Updated)* - Manages inventory within the warehouse.
6. **Purchase_Orders** *(PO_ID, Supplier_ID, Order_Date, Status, Expected_Delivery_Date, Total_Amount)* - Orders placed to suppliers for restocking inventory.
7. **Purchase_Order_Details** *(PO_Detail_ID, PO_ID, Product_ID, Quantity, Unit_Price)* - Itemized details of each purchase order.
8. **Sales_Orders** *(SO_ID, Customer_ID, Staff_ID, Order_Date, Status, Total_Amount, Subtotal_Amount, Discount_Amount, Shipping_Fee, Tax_Amount)* - Customer orders placed in the system.
9. **Sales_Order_Details** *(SO_Detail_ID, SO_ID, Product_ID, Quantity, Unit_Price)* - Itemized details of each sales order
10. **Shopping_Carts** *(Cart_ID, Customer_ID, Created_Date, Last_Modified, Total_Amount)* - Customer shopping carts for online/pending purchases.
11. **Category** *(Category_ID, manager_id, supplier_id, warehouse_id, Name, Description)* - Product categories for organization.

12. **Supplier_Categories** (Supplier_ID, Category_ID) - Links suppliers to the categories they provide.

13. **Payments** *(Payment_ID, SO_ID, Discount_ID, Amount, Payment_Date, Payment_Method, Payment_Status, Transaction_ID, Card_Last4, Card_Name, Expiry_Month, Expiry_Year)* - Tracks payment transactions for orders.

14. **Discount Management** (*Discount_ID, Name, Description, Start_Date, End_Date, Discount_Type, Discount_Value, Minimum_Purchase, Status*) – tracks the discount for customers' orders.

15. **Cart_Items** *(Cart_Item_ID, Cart_ID, Product_ID, Quantity)* - Individual items within shopping carts.

16. **Managers** *(Manager_ID, Manager_Name, Email, Phone_Number, Hire_Date, Salary, Is_Active, Age, Gender)* - Management personnel overseeing different aspects of the business.
17. **Sections** *(Section_ID, Section_Name, Manager_ID)* - Different departments/sections within the store.

Relationships:

1. **Managers → Sections:** One manager oversees multiple sections (1:M)
2. **Managers → Staff**: One manager supervises multiple staff members (1:M)
3. **Categories → Products**: One category contains multiple products (1:M)
4. **Categories → Warehouse**: One product can be stored in multiple warehouse locations (1:M)
5. **Customers → Sales_Orders**: One customer can place multiple sales orders (1:M)
6. **Customers → Shopping_Carts**: One customer can have one shopping cart (1:1)

7. **Staff → Sales_Orders**: One staff member can process multiple sales orders (1:M)
8. **Suppliers → Purchase_Orders**: One supplier can receive multiple purchase orders (1:M)
9. **Sales_Orders → Sales_Order_Details:** One sales order contains multiple line items (1:M)
10. **Sales_Orders → Payments**: One sales order can have one payment (1:1)
11. **Purchase_Orders → Purchase_Order_Details**: One purchase order contains multiple line items (1:M)
12. **Shopping_Carts → Cart_Items**: One cart contains multiple items (1:M)
13. **Products → Sales_Order_Details**: One product can appear in multiple order details (1:M)
14. **Products → Cart_Items**: One product can be in multiple cart items (1:M)
15. **Discount_Management → Payments**: One discount can be applied to multiple payments (1:M)
16. **Suppliers ↔ Categories**: Suppliers can provide multiple categories, and categories can be provided by multiple suppliers
17. **Sections → Staff**: Each staff can belong to one section, and each section can have multiple staffs (1:M)

Normalizations analysis:

1. Staff Table

Attributes: *(Staff_ID, Manager_ID, Name, Position, Section_ID, Salary, Hire_Date, Gender, Age, Assignment_count)*

1NF: SATISFIED

- All attributes contain atomic values
- No repeating groups
- Each staff member has a unique Staff_ID

2NF: SATISFIED

- Primary key is Staff_ID (single attribute)
- All non-key attributes (Manager_ID, Name, Position, Salary, Hire_Date, Gender) are fully dependent on Staff_ID
- No partial dependencies possible with single-attribute primary key

3NF: SATISFIED

Manager_ID is treated as a foreign key reference

2. Customers Table

Attributes: *(Customer_ID, Name, Email, Phone, Address, Birth_Date)*

1NF: SATISFIED

- All attributes are atomic
- No repeating groups
- Unique Customer_ID for each row

2NF: SATISFIED

- Single-attribute primary key (Customer_ID)
- All attributes fully dependent on Customer_ID

3NF: SATISFIED

- No transitive dependencies
- All non-key attributes depend directly on Customer_ID

3. Products Table

Attributes: *(Product_ID, Category_ID, Title, Price, Stock, Description, image_path)*

1NF: SATISFIED

- All attributes contain atomic values
- Unique Product_ID

2NF: SATISFIED

- Primary key is Product_ID (single attribute)
- All attributes fully dependent on Product_ID

3NF: SATISFIED

- Category_ID is a foreign key reference to Category table
- No transitive dependencies within this table

4. Suppliers Table

Attributes: *(Supplier_ID, manager_id, Name, Contact, Email, Address)*

1NF: SATISFIED

- Atomic values in all columns
- Unique Supplier_ID

2NF: SATISFIED

- Single primary key with full dependency

3NF: SATISFIED

- No transitive dependencies

5. Warehouse Table

Attributes: *(Warehouse_ID, Product_ID, Location_Name, Stock_Quantity, Last_Updated)*

1NF: SATISFIED

- All attributes are atomic
- Unique Warehouse_ID

2NF: SATISFIED

- Primary key is Warehouse_ID
- All attributes fully dependent on primary key

3NF: SATISFIED

- Product_ID is a foreign key reference
- No transitive dependencies

6. Purchase_Orders Table

Attributes: *(PO_ID, Supplier_ID, Order_Date, Status, Expected_Delivery_Date, Total_Amount)*

1NF: SATISFIED

- Atomic values throughout
- Unique PO_ID

2NF: SATISFIED

- Single primary key (PO_ID)
- Full functional dependency

3NF: SATISFIED

- Supplier_ID is foreign key reference
- No transitive dependencies

7. Purchase_Order_Details Table

Attributes: *(PO_Detail_ID, PO_ID, Product_ID, Quantity, Unit_Price)*

1NF: SATISFIED

- All attributes atomic
- Unique PO_Detail_ID

2NF: SATISFIED

- Primary key is PO_Detail_ID
- All attributes fully dependent on primary key

3NF: SATISFIED

- PO_ID and Product_ID are foreign key references
- No transitive dependencies

8. Sales_Orders Table

Attributes: *(SO_ID, Customer_ID, Staff_ID, Order_Date, Status, Total_Amount)*

1NF: SATISFIED

- Atomic values
- Unique SO_ID

2NF: SATISFIED

- Single primary key with full dependency

3NF: SATISFIED

- Customer_ID and Staff_ID are foreign key references
- No transitive dependencies

9. Sales_Order_Details Table

Attributes: *(SO_Detail_ID, SO_ID, Product_ID, Quantity, Unit_Price)*

1NF: SATISFIED

- All attributes atomic
- Unique SO_Detail_ID

2NF: SATISFIED

- Primary key is SO_Detail_ID
- Full functional dependency

3NF: SATISFIED

- SO_ID and Product_ID are foreign keys
- No transitive dependencies

10. Shopping_Carts Table

Attributes: *(Cart_ID, Customer_ID, Created_Date, Last_Modified)*

1NF: SATISFIED

- Atomic values
- Unique Cart_ID

2NF: SATISFIED

- Single primary key
- Full dependency

3NF: SATISFIED

- Customer_ID is foreign key
- No transitive dependencies

11. Category Table

Attributes: *(Category_ID, Name, Description)*

1NF: SATISFIED

- All attributes atomic
- Unique Category_ID

2NF: SATISFIED

- Single primary key
- Full functional dependency

3NF: SATISFIED

- No transitive dependencies

12. Supplier_Categories Table

Attributes: *(Supplier_ID, Category_ID)*

1NF: SATISFIED

- Atomic values
- Composite primary key (Supplier_ID, Category_ID)

2NF: SATISFIED

- Junction table with only foreign keys
- No non-key attributes to create partial dependencies

3NF: SATISFIED

- No non-key attributes
- No transitive dependencies possible

13. Payments Table

Attributes: *(Payment_ID, SO_ID, Discount_ID, Amount, Payment_Date, Payment_Method, Payment_Status, Transaction_ID)*

1NF: SATISFIED

- All attributes atomic
- Unique Payment_ID

2NF: SATISFIED

- Single primary key
- All attributes fully dependent on Payment_ID

3NF: SATISFIED

- SO_ID and Discount_ID are foreign key references
- No transitive dependencies

14. Discount_Management Table

Attributes: *(Discount_ID, Name, Description, Start_Date, End_Date, Discount_Type, Discount_Value, Minimum_Purchase, Status)*

1NF: SATISFIED

- Atomic values throughout
- Unique Discount_ID

2NF: SATISFIED

- Single primary key
- Full functional dependency

3NF: SATISFIED

- No transitive dependencies

15. Cart_Items Table

Attributes: *(Cart_Item_ID, Cart_ID, Product_ID, Quantity)*

1NF: SATISFIED

- All attributes atomic
- Unique Cart_Item_ID

2NF: SATISFIED

- Single primary key
- Full dependency

3NF: SATISFIED

- Cart_ID and Product_ID are foreign keys
- No transitive dependencies

16. Managers Table

Attributes: *(Manager_ID, Manager_Name, Email, Phone_Number, Hire_Date, Salary, Is_Active)*

1NF: SATISFIED

- All attributes atomic
- Unique Manager_ID

2NF: SATISFIED

- Single primary key
- Full functional dependency

3NF: SATISFIED

- No transitive dependencies

17. Sections Table

Attributes: *(Section_ID, Section_Name, Manager_ID)*

1NF: SATISFIED

- Atomic values
- Unique Section_ID

2NF: SATISFIED

- Single primary key
- Full dependency

3NF: SATISFIED

- Manager_ID is foreign key reference
- No transitive dependencies

Technologies Used:

To build the Cobra Shop database system, we will utilize the following technologies:

- Database Management System (DBMS): MySQL
- Backend Development: PHP, JavaScript
- Frontend Interface: HTML, CSS, JavaScript
- Hosting and Deployment: We will try to deploy our work on Google's free hosting cloud

Sample Queries:

1. Retrieve the total revenue generated by the store.
2. Retrieve all products that are currently out of stock.
3. Retrieve the names and emails of customers who have made a purchase in the last 6 months.
4. Retrieve the best-selling product overall.
5. Retrieve the total number of products sold per category.
6. Retrieve the most expensive product in the store.
7. Retrieve all customers who have spent more than $1000 in the store.
8. Retrieve the percentage of products in stock compared to total products.
9. Retrieve the most common payment method used by customers.
10. Retrieve the average number of products sold per order.
11. Retrieve all employees who were hired in the last year.

12. Retrieve the total amount of refunds processed.

13. Retrieve the salaries of all employes in 6 months.

14. Retrieve the percentage of employes gender.

15. Retrieve the number of visits for the website in 3 months.

16. Retrieve the category with the most visits.

17. Retrieve the percentage of ages of our customers.

18. Retrieve the percentage of costumers in different countries.

19. Retrieve the supplier who has provided the most products.

20. Retrieve all suppliers who have supplied products within the last 3 months.

21. Retrieve the best-selling product for each category.

22. Retrieve the most expensive product for each category.

23. Retrieve the total sales by week for the last 3 months.

24. Retrieve the total purchases by customer age groups (18-25, 26-40, 41-60, 60+).

25. Retrieve the number of customers and total sales from each country.

26. Retrieve the total sales amount processed by each staff member per month.

27. Retrieve the monthly revenue comparison between this year and last year.

28. Retrieve the top 5 customers with highest purchase frequency.

29. Retrieve the average time between order placement and delivery completion.

30. Retrieve the products with stock levels below 10% of their average monthly sales.

ERD Diagram:

We couldn't provide a picture because the diagram is too big to fit in, so we here's the link to ERD Diagram we created as a downloaded PNG image:

Link_to_PNG

Demo Video Link:

Link_to_Demo